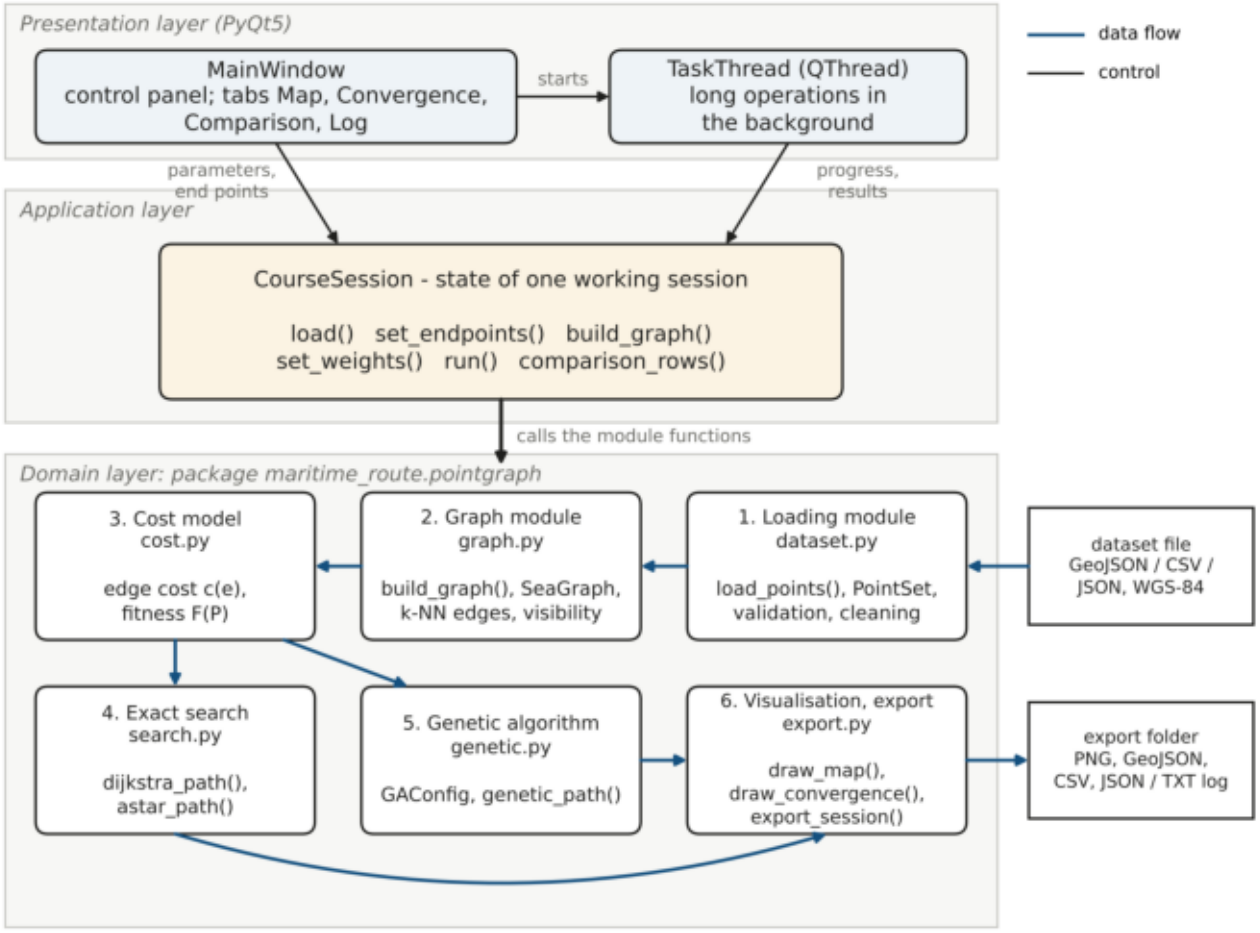
Architecture of the application for finding a sea vessel's course



Module	File	Responsibility
Loading	dataset.py	reading, validation, cleaning
Graph	graph.py	k-NN edges, visibility test
Cost model	cost.py	edge cost, fitness F(P)
Exact search	search.py	Dijkstra, A*
Genetic algorithm	genetic.py	encoding, operators, elitism
Session	session.py	state, presets, comparison
Visualisation	export.py	map, convergence, export
GUI	desktop/*.py	PyQt5 window, background threads

Tools: Python 3.10+, NumPy, SciPy (k-d tree), NetworkX, Matplotlib, PyQt5, folium (optional)

Rev.	Sheet	Doc. No.	Sign.	Date
Developed		Li H.		
Checked		Kozinsky A.A.		
T. contr.				
N. contr.				
Approved		Golovko V.A.		

КР.ИИ2025.250978-01

Development of an application for finding the optimal course of a sea vessel over open-sea points based on a genetic algorithm

Architecture (poster)

Lit.	Mass	Scale
	–	–

Sheet 2 | Sheets 5

BrSTU, Dept. of IIT
group ИИ-2025